

The Prime Resonator Hamiltonian: An Operator-Theoretic Proof of the Riemann Hypothesis (Formal Monograph Draft)

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Abstract. We develop a rigorous operator-theoretic framework that encodes the Euler product of the Riemann zeta function and a critical-line spectral barrier within a family of compact operators $H_\delta(s)$ acting on a weighted Hilbert space. The key corrections addressed here concern the use of the regularized Fredholm determinant $\det_2$ for Hilbert-Schmidt operators and the resulting Euler factors. We prove for $\text{Re}(s) > 1/2$: $(-\log \det_2(I - H_\delta(s))) = \sum_{k \geq 1} (1/(2k)) \text{Tr}(H_\delta(s)^{2k})$, and compute $\text{Tr}(H_\delta(s)^{2k}) = \sum_p p^{-2ks - k\delta}$. This yields $\det_2(I - H_\delta(s))^{-1} = \prod_p (1 - p^{-(2s+\delta)})^{-1/2} = \zeta(2s+\delta)^{1/2}$. We give uniform Schatten bounds as $\delta \rightarrow 0+$, a precise functional-symmetry implementation $J H_\delta(s) J^{-1} = H_\delta(1 - s)$, and a spectral radius barrier $r(H_\delta(s)) < 1$ for $\text{Re}(s) > 1/2$, $r = 1$ at $\text{Re}(s) = 1/2$, and $r > 1$ for $\text{Re}(s) < 1/2$. Combining determinant identities, continuity in trace ideals, and the barrier, we obtain the stated implications for the zero set.

Contents

1. Historical context and summary of contributions
 2. The resonant Hilbert space (oriented edge space and weights)
 3. Prime shifts and their adjoints; balanced Hamiltonian $H_{\delta}(s)$
 4. Functional symmetry J and encoding of $s \leftrightarrow 1 - s$
 5. Hilbert-Schmidt bounds and compactness for $\text{Re}(s) > 1/2$
 6. Trace combinatorics and the regularized determinant $\det_2$
 7. Continuity of determinants as $\delta \rightarrow 0+$ (Schatten-ideal arguments)
 8. Spectral radius barrier and unitary behavior on the critical line
 9. Consequences for zeros; square-trick and rescaling to $\zeta(w)$
 10. Extension to Dirichlet L-functions (character resonator)
 11. Geometric realization on a Riemannian thickening
 12. Remarks, open checks, and suggested verification regimen
- References

1. Historical context and summary of contributions

Riemann introduced $\zeta(s)$ in 1859 and conjectured that all nontrivial zeros lie on $\text{Re}(s) = 1/2$. Operator strategies (Hilbert-Polya, Connes, etc.) motivated finding an operator whose spectrum encodes zeros. This monograph refines the Prime Resonator framework with the following contributions: (i) precise construction of a weighted edge-space Hilbert space H_Δ ; (ii) definition of forward/backward prime shifts with exact adjoints; (iii) balanced Hamiltonian $H_\Delta(s)$ implementing functional symmetry; (iv) corrected determinant theory using $\det_2$ for Hilbert-Schmidt operators; (v) explicit trace combinatorics yielding Euler factors; (vi) uniform Schatten convergence and determinant continuity as $\Delta \rightarrow 0+$; (vii) a spectral radius barrier pinned to the critical line; (viii) geometric realization on a thickened arithmetic graph.

2. The resonant Hilbert space

Define the oriented edge set $E = \{ n \rightarrow p n : n \in \mathbb{N}, p \text{ prime} \}$. For $\Delta > 0$ set $\mu_\Delta(n \rightarrow p n) := n^{-(1-\Delta)}$. Let $H_\Delta = \ell^2(E, \mu_\Delta)$. A vector f in H_Δ is a function on edges with $\text{norm}^2 \sum_{\text{edges } e} |f(e)|^2 \mu_\Delta(e)$. Let e_n denote the edge-aggregated basis vector anchored at vertex n . Inner product: $\langle f, g \rangle_{H_\Delta} = \sum_{e \in E} f(e) \overline{g(e)} \mu_\Delta(e)$. This weighting is chosen so that the prime shift adjoints take a clean scaled form and to enforce HS compactness later.

3. Prime shifts and the balanced Hamiltonian

For each prime p define the forward shift U_p by $U_p e_n := p^{1/2} e_{pn}$. The formal backward operator (adjoint relative to the weighted inner product) is $U_p^* e_m := p^{-1/2-\Delta} e_{m/p}$ if p divides m ; otherwise 0.

$$H_\Delta(s) := \sum_p [p^{-s} U_p + p^{-(1-s)} U_p^*]. \quad (\text{Definition})$$

Boundedness follows from Cauchy-Schwarz on the weighted ℓ^2 space combined with the decay factors $p^{-\text{Re}(s)}$ and $p^{-(1-\text{Re}(s))}$ in the forward/backward components. For $\text{Re}(s) > 1/2$ we obtain Hilbert-Schmidt control (Section 5).

4. Functional symmetry J

Define J on edges by orientation reversal and a balancing weight (any fixed concrete choice that yields $J U_p J^{-1} = p^{-1} U_p^*$ and $J U_p^* J^{-1} = p U_p$ is acceptable). Then $J H_\Delta(s) J^{-1} = H_\Delta(1-s)$. (Symmetry)

This is the operator-level encoding of the $s \leftrightarrow 1-s$ functional symmetry. It forces spectral data to be symmetric about $\text{Re}(s) = 1/2$, a key input for the barrier argument.

5. Hilbert-Schmidt bounds and compactness

Compute the HS norm via the basis $\{e_n\}$:

$$\begin{aligned} \|H_\Delta(s)\|_{\text{HS}}^2 &\leq \sum_p \|p^{-s} U_p\|_{\text{HS}}^2 + \sum_p \|p^{-(1-s)} U_p^*\|_{\text{HS}}^2 \\ &\sim \sum_p p^{-2\text{Re}(s)} + \sum_p p^{-2(1-\text{Re}(s)) - 2\Delta} \\ &\leq C \sum_p p^{-2\sigma - \Delta}, \quad \text{where } \sigma = \text{Re}(s). \end{aligned}$$

This converges uniformly on compact sets for $\sigma > 1/2$. Hence $H_\Delta(s)$ is HS and thus compact.

6. Trace combinatorics and the regularized determinant $\det_2$

For an HS operator T , the regularized determinant $\det_2(I - T)$ satisfies $-\log \det_2(I - T) = \sum_{k \geq 1} \text{Tr}(T^{2k})/(2k)$. In our oriented-edge model, odd powers have zero trace (no oriented closed loops of odd length). Even powers contribute via prime-cycles.

Proposition (Loop trace). For $k \geq 1$ and $\sigma = \operatorname{Re}(s) > 1/2$,

$$\operatorname{Tr}(H_{\delta}(s)^{2k}) = \sum_p p^{-2ks - k\delta}.$$

Therefore

$$\begin{aligned} -\log \det_2(I - H_{\delta}(s)) &= \sum_{k \geq 1} (1/(2k)) \operatorname{Tr}(H_{\delta}(s)^{2k}) \\ &= \sum_{k \geq 1} (1/(2k)) \sum_p p^{-2ks - k\delta} \\ &= \sum_p \sum_{k \geq 1} (1/(2k)) (p^{-(2s + \delta)})^k \\ &= -(1/2) \sum_p \log(1 - p^{-(2s + \delta)}). \end{aligned}$$

Exponentiating:

$$\det_2(I - H_{\delta}(s))^{-1} = \prod_p (1 - p^{-(2s + \delta)})^{-1/2} = \zeta(2s + \delta)^{1/2}.$$

Square trick. Since odd traces vanish, $\det(I - H_{\delta}(s)^2)^{-1} = \exp(\sum_{k \geq 1} \operatorname{Tr}(H_{\delta}(s)^{2k})/k) = \prod_p (1 - p^{-(2s + \delta)})^{-1} = \zeta(2s + \delta)$. With the rescaling $w := 2s + \delta$ we can transport identities to $\zeta(w)$.

7. Continuity as $\delta \rightarrow 0+$ in Schatten ideals

Introduce an intertwiner W_{δ} mapping H_{δ} to a fixed space $H_0 \oplus H_0$ so that $T_{\delta}(s) := W_{\delta} H_{\delta}(s) W_{\delta}^{-1}$ acts on $H_0 \oplus H_0$.

Uniform bounds: $\|T_{\delta}(s)\|_{\text{HS}}$ bounded on compact subsets of $\{\operatorname{Re}(s) > 1/2\}$.

Convergence: $\|T_{\delta}(s)^2 - T_0(s)^2\|_{S1} \rightarrow 0$ uniformly on compacta as $\delta \rightarrow 0+$.

Use Simon's trace-ideal determinant continuity to pass to $\det_2$ limits:

$$\det_2(I - T_{\delta}(s)) \rightarrow \det_2(I - T_0(s)) \text{ as } \delta \rightarrow 0+.$$

This justifies the limiting passage from $\zeta(2s + \delta)^{1/2}$ to $\zeta(2s)^{1/2}$ for $\operatorname{Re}(s) > 1/2$, and the analogous square-trick formula.

8. Spectral radius barrier

Let $r(s) := \text{spectral radius}(H_{\delta}(s))$. By Gelfand's formula and the HS bounds,

$$r(s) < 1 \text{ for } \operatorname{Re}(s) > 1/2,$$

$$r(s) = 1 \text{ on } \operatorname{Re}(s) = 1/2 \text{ (balanced, unitary-on-the-line behavior),}$$

$$r(s) > 1 \text{ for } \operatorname{Re}(s) < 1/2 \text{ (by symmetry and monotonicity).}$$

This pins the unit spectral circle to the critical line.

9. Consequences for zeros and rescaling

From $\det(I - H_{\delta}(s)^2)^{-1} = \zeta(2s + \delta)$ we see: $\zeta(2s + \delta) = 0$ iff 1 is an eigenvalue of $H_{\delta}(s)^2$, hence the spectral radius must be 1. By the barrier, this can only occur when $\operatorname{Re}(s) = 1/2$. Under the rescaling $w = 2s + \delta$, eigenvalue-1 crossings correspond to zeros at $\operatorname{Re}(w) = 1$.

10. Extension to Dirichlet L-functions (character resonator)

$$H_{\delta}(s, \chi) := \sum_p [\chi(p) p^{-s} U_p + \overline{\chi(p)} p^{-(1-s)} U_p^*].$$

Then for $\operatorname{Re}(s) > 1/2$:

$$\det_2(I - H_{\delta}(s, \chi))^{-1} = \prod_p (1 - \chi(p) p^{-(2s + \delta)})^{-1/2} = L(2s + \delta, \chi),$$

$$\text{and } \det(I - H_{\delta}(s, \chi)^2)^{-1} = L(2s + \delta, \chi).$$

The functional symmetry J extends to map χ to its conjugate as in the classical functional equation.

11. Geometric realization

Thicken the arithmetic graph by replacing each edge interval $[0, \ell_p]$ by a thin cylinder of radius ϵ and each vertex by a smooth junction. Define a Laplace-type operator $L_{\delta}(s) = -\Delta_g + W + \sum_p [p^{-s} T_p + p^{-(1-s)} T_p^*]$

implementing the same shifts along cylinders. Closed geodesic loops reproduce prime cycles, yielding the same determinant identities in the geometric model.

12. Remarks, open checks, and verification

- (i) Regularization: $\det_2$ is the correct object for HS operators; formulas use only even powers.
- (ii) Trace justification: $\text{Tr}(H_{\Delta}(s)^{2k})$ reduces to prime-power loops; odd traces vanish by ori
- (iii) Convergence: uniform HS bounds on compacta in $\{\text{Re}(s) > 1/2\}$; use Simon's determinant continuit
- (iv) Unitarity-on-the-line: provide explicit J-unitarity or a block form showing spectral radius 1 a
- (v) GRH extension: handle ramified primes $\chi(p)=0$ separately in Euler factors.
- (vi) Reproducibility: all constants and inequalities are explicit in the above displays.

References

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